

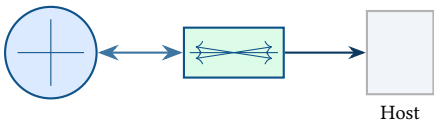
EclipseStack Technical Report

Architecture, Tooling, and Implementation Notes

Simon Knight
Adelaide, Australia

March 2026 • Version 0.1.0

Last updated: March 11, 2026



Abstract. This technical report documents EclipseStack’s architecture, build and release workflow, and implementation decisions. It is intended as a living engineering reference that captures system boundaries, key components, and rationale for design choices.

Contents

List of Design Decisions & Rules	2
1 Overview	3
2 System Architecture	3
3 Build, Test, and Release	3
4 Design Decisions	4
List of Figures	5
List of Tables	5

List of Design Decisions & Rules

1	Scope	3
2	Repository Layout	3
1	Design Rule: Reproducible Builds	4
3	Example Decision	4

1 Overview

This report follows the sk-techreport format from `/dev/papers`. Use it to record architectural intent and the reasoning behind changes.

Design Decision 1: Scope

This document focuses on: system architecture, module boundaries, build/test tooling, and design decisions that impact maintainability.

2 System Architecture

Describe the high-level architecture here, including the main subsystems and their interfaces.

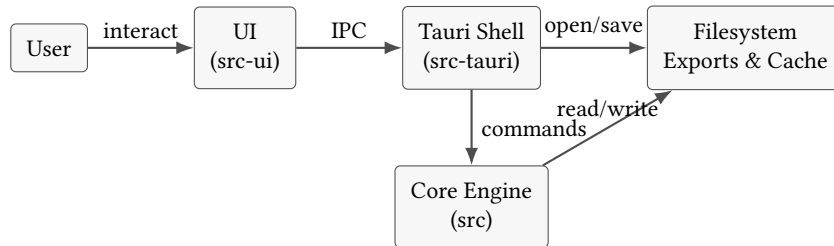


Figure 1. High-level component boundaries and data flow.

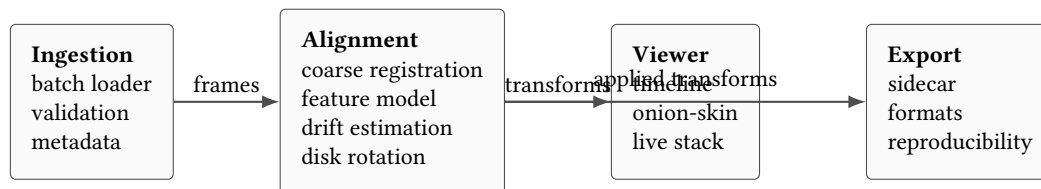


Figure 2. Core processing pipeline from ingestion to export.

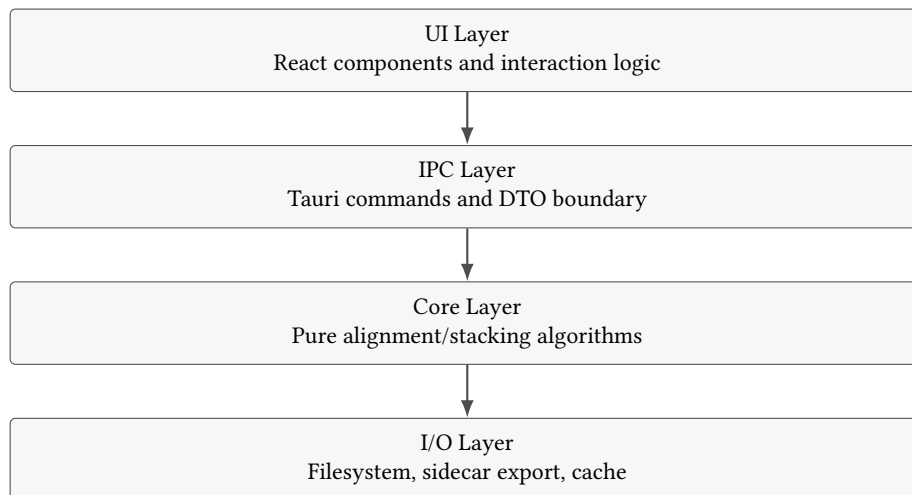


Figure 3. Layering model: keep algorithms independent from UI concerns.

Design Decision 2: Repository Layout

Record why directories such as `src`, `src-tauri`, and `src-ui` exist, and how responsibilities are split.

3 Build, Test, and Release

Document build commands, CI expectations, and release steps.



Figure 4. Typical developer and CI pipeline for changes.



Figure 5. Change management path from local edits to release artifacts.

: Reproducible Builds

Builds should be reproducible on developer machines and CI by pinning toolchains and documenting required dependencies.

4 Design Decisions

Add design decisions as they are made. Keep each one small and focused.

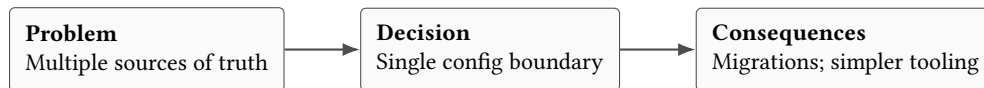


Figure 6. Decision record structure: problem, decision, consequences.

Design Decision 3: Example Decision

Decision: Choose a single source of truth for configuration.

Rationale: Avoid divergence between multiple config formats.

Consequences: Some migrations may be required.

List of Figures

1	High-level component boundaries and data flow.	3
2	Core processing pipeline from ingestion to export.	3
3	Layering model: keep algorithms independent from UI concerns.	3
4	Typical developer and CI pipeline for changes.	4
5	Change management path from local edits to release artifacts.	4
6	Decision record structure: problem, decision, consequences.	4

List of Tables
